

Python Notes by AVD Group

Arrays --> Collection of same type of elements are called as Array

Store student marks ---> 99, 100, 95, 90

Store Email id's ---> 'venkat@ibm.com', 'varma@tcs.com'

Store height ---> 4.8, 5.0, 4.0, 5.8, 6.2

--> To work for Arrays we have 3 option in python

1. List
2. array module
3. numpy.array()

List : If you want to store group of elements then use list

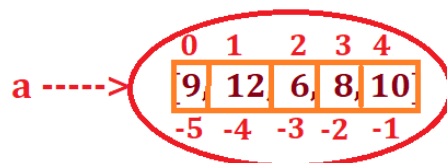
In list we can store same type or different type of data also

Ex: marks = [95, 99, 100, 98]

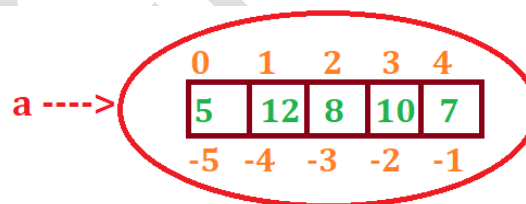
emp1 = [101, 'Venkat', 25000.0]

a = [9, 12, 6, 8, 10]

To know total no.of elements : len(a)



a = [5, 12, 8, 10, 7]



--> Display Array Values

1. Display all elements at a time by using array name

print(a)

2. Display all elements one by one by using for loop

for i in a:
print(i)

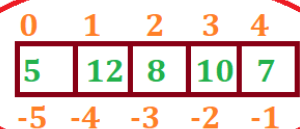
3. Display specific element by using index number

print(a[-3]) ---> 8

print(a[4]) ---> 7

print(a[5]) ---> Index Error

a ---->



4. Display part of data by using slicing operation

print(a[1:4]) ---> 12 8 10

print(a[1:7]) ---> 12 8 10 7

print(a[::]) ---> 5 12 8 10 7

print(a[::-1]) ---> 7 10 8 12 5

print(a[4:1]) ---> []

a[4:1:1] --> a[4] a[5] a[6] a[7]

a[4:1:-1] --> a[4] a[3] a[2]

Python Notes by AVD Group

--> Consider a = [10, 15, 12, 6, 9], Now Display only Even Number

```
a = [10, 15, 12, 6, 9]
for i in a:
    if i%2==0:
        print(i, end='\t')
```

	10	12	6	9
i	10	15	12	6
	10%2==0 ✓	15%2==0 ✗	12%2==0 ✓	6%2==0 ✓

--> Consider a = [10, 15, 12, 6, 9], print all elements one by one by using index number

a = [10, 15, 12, 6, 9]

```
for i in a:
    print(i)
```

i
10

```
print(a[0])
print(a[1])
print(a[2])
print(a[3])
print(a[4])
```

```
for i in range(0,5):
    print(a[i])
```

i	0	1	2	3	4
	10	15	12	6	9

--> Consider a = [10, 15, 12, 6, 9]

Now read an element then check element is available or not,
if available then print index number of that element

input: 12

Output: 2

input : 20

Output: Element not Available

```
a = [10, 15, 12, 6, 9]
n = int(input('Enter any Element : '))
for i in range(len(a)):
    if a[i]==n:
        print(f'{n} available in index no {i}')
        break
    else:
        print(f'{n} not available in your array')
```

n	i	
15	0	10==15 ✗
	1	15==15 ✓

--> Consider a = [10, 15, 12, 6, 9]

To modify :

a[index] = value

Now Replace Element --> Read an element if available replace with new element

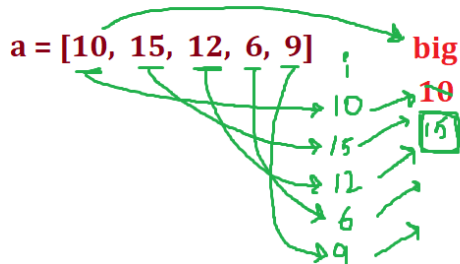
input : Enter element to change : 15
Enter new Element : 20
Your Array is [10, 20, 12, 6, 9]
a[1] = 20

input : Enter element to change : 14
14 is not available
Your Array is [10, 20, 12, 6, 9]

- > Check element is available or not --> if n in a:
- > if available then find index number
- > Then replace with new element

```
a = [10, 15, 12, 6, 9]
n = int(input('Enter any Element : '))
if n in a:
    m = int(input('Enter new number'))
    for i in range(len(a)):
        if a[i]==n:
            print(f'{n} available in index no {i}')
            a[i] = m
            break
else:
    print(f'{n} not available in your array')
print(a)
```

--> Consider a = [10, 15, 12, 6, 9] , now find biggest number



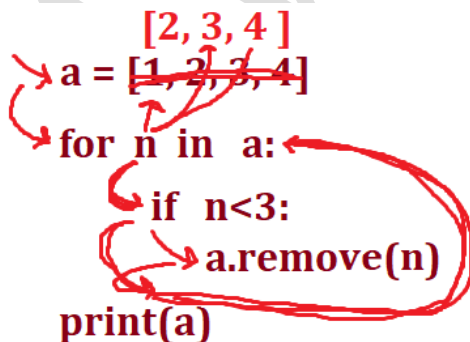
`print(max(a))` ×

`max()`

```
a = [10, 15, 12, 26, 9]
maxindex = 0
for i in range(len(a)):
    if a[i]>a[maxindex]:
        maxindex = i
print(f'Biggest Number is {a[maxindex]}')
```

```
a = [10, 15, 12, 26, 9]
max = a[0]
for i in a:
    if i>max:
        max = i
```

```
print(f'Biggest Number is {max}')
```



`n`
`*`
`8`
`4`